

Networking Protocols — Detailed Lecture Notes

JKSSB Computer Awareness

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41.1 Why protocols and addresses matter

Computers on a network can understand each other only because they follow agreed rules. A **protocol** is such a set of rules: it fixes what messages look like and in which order the machines exchange them. Different jobs need different protocols. Opening a web page, moving a file, sending an email, translating a name into an address, and handing out addresses to new devices are five different jobs, so they use five different protocols. The exam tests each protocol by its **exact English name** together with the single role it performs.

Definition (Protocol). *A fixed set of rules governing communication between devices on a network, covering message format and exchange order. TCP, IP, DNS, DHCP, HTTP, HTTPS, FTP, Telnet, and SMTP are the protocols of this lesson.*

Term	Meaning in this lesson
TCP	Transmission Control Protocol: reliable, ordered, connection-oriented delivery.
UDP	User Datagram Protocol: fast, connectionless, best-effort delivery.
IP	Internet Protocol: addressed, best-effort carriage of packets across networks.
ICMP	Internet Control Message Protocol: control and diagnostic messages only.
DNS	Domain Name System: translates names into IP addresses.
DHCP	Dynamic Host Configuration Protocol: automatically assigns addresses.
HTTP / HTTPS	HyperText Transfer Protocol (plain) and its Secure version for the web.
FTP	File Transfer Protocol: moves files between computers.
Telnet	Plain-text remote login protocol.
SMTP	Simple Mail Transfer Protocol: sends outgoing mail.
OSI model	The seven-layer reference model from Physical to Application.
Port	A number naming the service on a machine (HTTP 80, HTTPS 443).

The rest of this lesson uses these terms in one consistent sense. **TCP** and **UDP** are the two transport protocols. **IP** names the addressing protocol with its two versions, IPv4 and IPv6. **ICMP** is the helper that reports errors, never the carrier of user data. **DNS** and **DHCP** are the two address services: one resolves names, the other assigns addresses. **HTTP**, **HTTPS**, **FTP**, **Telnet**, and **SMTP** are application protocols, each with one job and one well-known port.

41.2 TCP and UDP

The transport layer offers two opposite bargains. **TCP** (Transmission Control Protocol) is connection-oriented: it sets up the connection, numbers every segment, retransmits whatever is lost, and delivers the bytes complete and in order. What TCP guarantees is *ordering and correctness*; it does not promise when the data will arrive. **UDP** (User Datagram Protocol) is connectionless: it sends each datagram independently, with no setup, no ordering, and no retransmission. What UDP offers in return is speed and low overhead.

Definition (TCP). *A connection-oriented transport protocol that delivers data reliably and in order through sequencing, acknowledgement, and retransmission.*

Definition (UDP). *A connectionless transport protocol that sends independent datagrams quickly with no delivery, ordering, or error-recovery guarantee.*

Choose between them by what the task values most. A file, a web page, or an email must arrive complete and correct, so it travels over TCP. Live voice, a video stream, or a quick DNS lookup values speed over perfection, so it travels over UDP, where a lost fragment is simply skipped.

Example 41.2.1. *An item states, “TCP guarantees fast delivery of every packet.” This is wrong. TCP guarantees that the bytes arrive complete and in order (PYQ-18); speed is exactly what TCP sacrifices. The fast, best-effort protocol is UDP.*

41.3 IPv4 and IPv6

The **Internet Protocol (IP)** carries addressed packets from the sender to the receiver across networks. Each device holds an IP address, and routers read that address to forward the packet hop by hop. IP by itself is unreliable: delivery is best-effort, and it is TCP above it that adds reliability.

IPv4 uses 32-bit addresses written as four decimal octets, for example 192.168.1.10. Its header is variable in length (20 to 60 bytes). **IPv6** uses 128-bit addresses written as eight hexadecimal groups, for example 2001:0db8::1, and its header is fixed at 40 bytes (PYQ-18, TRM-10). The longer address space is the whole point of IPv6: it supplies vastly more addresses than IPv4.

Definition (IPv4). *The 32-bit version of the Internet Protocol, with dotted-decimal addresses and a variable-length header.*

Definition (IPv6). *The 128-bit version of the Internet Protocol, with hexadecimal addresses and a fixed 40-byte header.*

41.4 ICMP: the control channel

ICMP (Internet Control Message Protocol) carries control and diagnostic messages about the network itself: error reports such as “destination unreachable” and reachability probes. Its two familiar tools are **ping**, which asks whether a host is reachable, and **traceroute**, which shows the path packets take. ICMP transfers no application data and uses no port numbers (TRAP-19). That last point is the exam’s favourite trap: any option giving ICMP a port, a file transfer, or a mail delivery is wrong.

41.5 DNS and DHCP

DNS (Domain Name System) translates a human name such as `jkssb.nic.in` into the IP address the network needs. Without it every user would have to memorise numeric addresses. DNS queries travel mainly over UDP on port 53, falling back to TCP for large zone transfers.

DHCP (Dynamic Host Configuration Protocol) automatically assigns an IP address, subnet mask, default gateway, and DNS server address to each device that joins the network. Without it every address would have to be typed in by hand. The home router and the office network each run a DHCP server; the joining laptop or phone acts as the DHCP client, using UDP ports 67 and 68.

Definition (DNS). *The distributed naming service that resolves a domain name into its IP address. It answers “what number goes with this name?”*

Definition (DHCP). *The service that automatically assigns IP configuration (address, mask, gateway, DNS server) to devices joining a network. It answers “what address may this newcomer use?”*

Example 41.5.1. *An item asks which service gives a newly connected office desktop its IP address. Trace the pair: DNS resolves names the user already has, while DHCP hands out addresses to newcomers. The answer is DHCP. If the stem instead asks what turns `jkssb.nic.in` into an address, the answer is DNS.*

41.6 Application protocols: web, files, remote login, mail

HTTP (HyperText Transfer Protocol, port 80) fetches web pages in plain text. **HTTPS** (HyperText Transfer Protocol Secure, port 443) is the same protocol with an SSL/TLS encryption layer, shown by `https://` and the padlock. The S stands for Secure (WO 2026 Q76 territory).

FTP (File Transfer Protocol, port 21) moves files between computers; it is neither email nor web browsing. **Telnet** (port 23) opens a plain-text remote login session and is today replaced by the encrypted SSH. **SMTP** (Simple Mail Transfer Protocol, port 25) sends outgoing mail to the server. Receiving is a separate job handled by **POP3** (Post Office Protocol version 3, port 110), which downloads mail to the local machine (PYQ-16, TRAP-17), and **IMAP** (Internet Message Access Protocol, port 143), which keeps mail synced on the server.

41.7 The seven OSI layers

The **OSI model** (Open Systems Interconnection) is a seven-layer reference frame for describing network communication:

No.	Layer	Job
7	Application	Services the user sees: HTTP, FTP, SMTP, DNS, DHCP.
6	Presentation	Translation, encryption, compression.
5	Session	Opens, manages, and closes sessions between machines.
4	Transport	End-to-end delivery: TCP and UDP.
3	Network	Addressing and routing: IP (IPv4, IPv6) and ICMP.
2	Data Link	Frames on a single link: switch, MAC address.
1	Physical	Bits on the wire, cable, or radio signal.

Data moves down the layers on sending and up the layers on receiving, each layer adding (or reading) its own header. For this exam, three placements carry almost all the marks: the application names (HTTP, FTP, SMTP, DNS, DHCP) at layer 7, TCP and UDP at layer 4, and IP and ICMP at layer 3.

Example 41.7.1. *Opening a web page uses HTTP at the Application layer, TCP at the Transport layer, and IP at the Network layer: one protocol per layer, working together. An item asking “which layer does IP belong to” is answered by the same table: the Network layer (layer 3).*

41.8 Port awareness

An **IP address** names the machine; a **port number** names the service on that machine. Ports 0–1023 are the well-known ports. The exam set is: FTP 21, Telnet 23, SMTP 25, DNS 53, HTTP 80, POP3 110, IMAP 143, HTTPS 443. DHCP uses UDP ports 67 and 68, and ICMP uses no ports at all. The recommended drill is to cover the service names and recall the numbers in the chain 21–23–25–53–80–443.

41.9 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. IPv6 is **128-bit**, never 256-bit; its header is fixed at 40 bytes (TRAP-18, TRM-10).
2. ICMP is **control/diagnostic** only: no application data, no ports (TRAP-19).
3. TCP promises **order**, not timing; UDP promises neither (PYQ-18).
4. **SMTP** sends mail; **POP3/IMAP** receive it (TRAP-17).
5. **DNS** resolves names to addresses; **DHCP** assigns addresses to joining devices.
6. HTTP is port **80** and plain; HTTPS is port **443** with SSL/TLS.
7. TCP and UDP live at the **Transport** layer; IP and ICMP at the **Network** layer; HTTP, FTP, SMTP, DNS, and DHCP at the **Application** layer.

Exercises

1. Distinguish TCP from UDP on reliability, connection setup, and one typical use each.
2. Contrast IPv4 with IPv6 on address length, notation, and header size.
3. Explain why ICMP is described as a control protocol and name its two common tools.
4. Distinguish DNS from DHCP with one sentence on each role.
5. List the seven OSI layers in order with the protocols of this lesson placed at their layers.

Solutions

1. TCP is connection-oriented, reliable, and ordered: it numbers segments and retransmits losses; typical uses are web pages, email, and file transfer. UDP is connectionless and best-effort: no setup, ordering, or retransmission; typical uses are streaming, voice calls, and DNS queries.
2. IPv4 is 32-bit, dotted-decimal (four octets), with a variable 20–60 byte header. IPv6 is 128-bit, hexadecimal (eight groups), with a fixed 40-byte header.
3. ICMP carries control and diagnostic messages about the network (errors, reachability) rather than user data, and it uses no ports. Its tools are ping (reachability) and traceroute (path).
4. DNS translates a domain name into its IP address. DHCP automatically assigns an IP address and related configuration to a device joining the network.
5. 7 Application (HTTP, HTTPS, FTP, Telnet, SMTP, POP3, IMAP, DNS, DHCP); 6 Presentation; 5 Session; 4 Transport (TCP, UDP); 3 Network (IP, ICMP); 2 Data Link; 1 Physical.